Project 1: Competitive Analysis

Erdős Institute Spring 2026 UX Research
Dejan Duric, Tamera Jones, Ibrahim Odugbemi, David Stifler

Scenario

Our company, **Kairos**, offers remote patient monitoring (RPM) devices and software for managing chronic diseases such as diabetes and heart failure. RPM is growing rapidly, but patient adherence and integration with healthcare providers remain significant challenges. Our leadership has asked us to assess the **current state of our app** and suggest **future product development directions** based on projected profit and loss derived from our research.

Outline of Tasks

1. We conducted real **market research** based on our scenario. Where necessary, we supplemented this real data with synthetic data. We carried out a **SWOT analysis**, including detailed comparisons between our app and competitors, ensuring that each element was connected to profit or loss with supporting data. We also conducted additional background research required to make an informed pitch.
2. We wrote an **executive summary** of our analysis for the Chief Product Officer to determine how our company could best position its RPM solution against competitors. We provided clear recommendations for future development.
3. We developed a **case study** designed for potential customers (patients), aimed at convincing them that our app was the best option for their needs.

KAIROS

Hypothetical Company Profile

Company Overview

- **Founded:** 2023.
- **Headquarters:** Little Rock, Arkansas (Strategic proximity to high-need rural regions).
- **Founders/CEO:** Dr. Elena Pappas (CEO) and Marcus Thorne (CTO).
- **Origin:** Founded to address the "wait-time crisis" in rural Arkansas and the Mississippi Delta. The name **Kairos** (καιρός) reflects the "critical window" for intervention in preeclampsia and postpartum hemorrhage, which are leading causes of maternal death in noncore areas.
- **Mission:** "Essential Care for the Isolated," focusing on providing clinical-grade monitoring to the 2.3 million women in maternity care deserts who skip prenatal visits due to distance or cost.
- **Website:** kairos-rural.io (Simulated).

Funding History (Plausible/Synthetic Data)

- **Total Raised:** $8.2 million.
- **Funding Timeline:**
  - **2023 (Oct):** $1.2M Seed Round led by Delta Regional Venture Partners.
  - **2025 (Jan):** $7M Series A focused on **RHTP Integration**, with participating investors from the Arkansas Rural Health Partnership and Federal Equity Grants.
- **Strategic Alignment:** Currently pursuing a portion of the $17M CMS Transforming Maternal Health (TMAH) grants awarded to 15 states to test scalable care models.

The Product Redesign: Kairos "Desert Shield" Platform

Kairos is an ultra-low-bandwidth virtual care platform specifically for rural and frontier medical environments.

1. **"Kairos Core" App:** A lightweight Progressive Web App (PWA) under 5MB. It uses localized SMS-based data syncing for patients without 5G/LTE access.
2. **RHTP Reporting Dashboard:** A provider interface that maps patient outcomes directly to **Rural Health Transformation Program** strategic goals (Preventive Care, Innovative Care Models, and Sustainable Access).

3. **Kairos Cellular Bridge:** A proprietary, low-power cellular gateway that connects standard Bluetooth cuffs and scales directly to the cloud without requiring a patient

smartphone or home WiFi.

4. **Community-Led Care Layer:** Unlike Marani's MFM-led model, Kairos trains and integrates **local community health workers (CHWs)** and "Patient Navigators" who already exist in these communities, addressing the physician shortage.

Proprietary Hardware / Devices

- **The "Bridge" Gateway:** A ruggedized, solar-rechargeable cellular hub that serves as the "hotspot" for the patient's vitals. This removes the "Smartphone Requirement" barrier common in Babyscripts and Tenovi platforms.
- **Clinical Moat:** Integration of "Offline Risk-Scoring" (ORS) which performs risk stratification on the device itself before transmission, ensuring alerts are prioritized before the data even reaches a congested cell tower.

AI / Risk Stratification Engine

- **Kairos Predict:** A "Minimalist AI" model that focuses on the three highest-risk indicators in rural populations: blood pressure spikes, gestational weight gain, and mental health "red flags" through the use of expert-designed questionnaires and check-ins.
- **Data Advantage:** While smaller than Babyscripts, Kairos holds the largest dataset for **low-bandwidth patient behavior**, allowing for unique insights into adherence in isolated environments.

Go-to-Market: Strategic Segments

- **Rural Health Clinics (RHCs) & FQHCs:** Targets the 52.4% of rural hospitals that no longer offer obstetric services. Kairos acts as a "Virtual OB Unit" for these general clinics.
- **State Medicaid Agencies:** Leveraging the fact that 47% of rural births are covered by Medicaid, Kairos is positioned as a tool to reduce the higher mortality rate seen in rural Medicaid populations. For example, rural residents face 9% higher probability of severe maternal morbidity and mortality compared to urban residents. These disparities compound with racial inequities: Black women face 3.5x the maternal mortality rate of White women nationally. State-level data suggest these gaps widen further in rural settings: in rural Georgia, Black women face double the maternal mortality rate of rural White women.
- **RHTP Grant Recipients:** Direct partnership with health systems receiving RHTP funds to provide the "Tech Innovation" requirement of their grant.

Key Data Points & Projected Impact (Synthetic/Plausible)

- **Adherence Improvement:** Following the UX redesign for low-signal areas, projected patient adherence is expected to rise from **42% to 78%**.
- **Data Reliability:** Achieved 98% data transmission success in "Edge Network" environments (zones with <1 bar of LTE signal).

- **Distance Mitigation:** Average reduction in "Maternal Travel Miles" projected at **450 miles** per pregnancy episode.

ROI & Financial Framework

- KAIROS delivers 41–113% ROI for rural clinics through CPT reimbursement alone at 50–80 enrolled patients, **with break-even at 54 patients**. When factoring in emergency transport avoidance — where a single prevented air transfer saves $45,000 — clinic ROI exceeds 400%. RHTP grant alignment further reduces clinic out-of-pocket costs by 30–50%.
- **RHTP Sustainability:** By aligning with RHTP, Kairos provides an estimated **145% ROI** for rural clinics through a combination of **CPT code capture (99453/99454)** and "Transformation Incentives".
- **Cost Avoidance:** A single emergency air transport from a rural county to an urban Level III facility averages $36,000–$60,000 depending on distance (American Action Forum, 2023; CMS rural rate multiplier of 1.5x)

- **Formula:** $ROI = \frac{TotalSavings + NewRevenue - ImplementationCost}{ImplementationCost} \times 100$

Strengths

- **Broadband-Agnostic:** Only competitor optimized specifically for "Broadband Deserts" using SMS-based information.
- **RHTP Optimization:** The only platform with a built-in grant-reporting engine for rural federal funding.
- **Low Barrier to Entry:** Does not require smartphone ownership or EMR expertise.

Weaknesses / Gaps

- **Brand Awareness:** Lacks the national "incumbent" status of Babyscripts, making providers and patients less likely to recommend or seek out our products.
- **Feature Set:** Purposely lacks "Concierge" features (video yoga, lactation 1-on-1s) to remain lightweight, which may deter high-end commercial payers.
- **Scaling Speed:** Limited by the slow rollout of federal rural health infrastructure.

Opportunities

- **Underserved Population:** 35%+ of U.S. counties are classified as “maternity care deserts;” a large population needs care they are not getting through existing channels.
- **Untapped Market:** Existing RPM systems provide premium service at a premium price; we focus on essential services at a price point affordable to everyone.
- **Improvement Potential:** Overall D+ score on preterm births in the U.S. indicates that any new medical intervention has the capacity to move the needle in the right direction.

Threats

- **Technology Barrier:** Communities lacking access to technology are less able to adapt RPM of any kind; our target market is the most challenging one to reach in the first place.
- **Limited Growth Potential:** While underserved (esp. rural) communities are not seeing the same decline in birth rate that urban communities are, they are also not showing growth; we may see market saturation sooner than we would like for our RPM solution.
- **Political Volatility:** Erratic and unpredictable executive branch of US government liable to change stance rapidly and without warning, jeopardizing RHTP and other policies.

KAIROS Per-Clinic ROI Model (Year 1)

Clinic perspective: what the clinic pays out of pocket vs. what it gets back

	Conservative	Base Case	Optimistic	Source / Notes
ASSUMPTIONS				
Patients enrolled per year	30	50	80	*Synthetic; small rural clinic / FQHC*
Monitoring duration (months per patient)	3	3	3	*KAIROS 12-week postpartum window*
Emergency transports prevented per year	0	1	2	*Synthetic; early detection of preeclampsia / PPH*
Cost per emergency air transport	$45,000	$45,000	$45,000	*AAF 2023 ($36K median); CMS rural 1.5x multiplier*
RHTP grant offset (% of clinic costs covered)	0%	30%	50%	*Synthetic; partial RHTP grant coverage of tech costs*
WHAT THE CLINIC PAYS				
KAIROS platform license ($35/patient/month)	$3,150	$5,250	$8,400	*Low end of industry range ($30–$80); rural pricing*
Devices: BP cuff + scale + Bridge (net $120/patient)	$3,600	$6,000	$9,600	*Cuff ~$60, scale ~$40, Bridge ~$100; ~40% reused*
CHW / Navigator time ($25/hr × 1.5 hrs/patient)	$1,125	$1,875	$3,000	*BLS median CHW wage ~$22–$25/hr*
Training & onboarding (one-time, Year 1)	$3,000	$3,000	$3,000	*Synthetic estimate*
Gross Cost (before grant)	**$10,875**	**$16,125**	**$24,000**	
Less: RHTP grant offset	$0	$4,838	$12,000	*Grant covers portion of implementation*
NET COST TO CLINIC (out of pocket)	**$10,875**	**$11,288**	**$12,000**	
WHAT THE CLINIC GETS BACK				
CPT 99453: Device setup ($22 × patients)	$660	$1,100	$1,760	*Tenovi 2026 RPM CPT Guide; CMS 2026 PFS*
CPT 99454: Device supply + data ($47/patient/month)	$4,230	$7,050	$11,280	*CMS 2026 proposed rate: $47.06*
CPT 99457: Clinical management ($52/patient/month)	$4,680	$7,800	$12,480	*Prevounce 2026 billing guide*
Total CPT Reimbursement	**$9,570**	**$15,950**	**$25,520**	
Cost avoidance: prevented	$0	$45,000	$90,000	*Per event: $45K (AAF / CMS rural)*

emergency transports				
Total gains WITH cost avoidance	**$9,570**	**$60,950**	**$115,520**	
CLINIC ROI				
ROI — CPT reimbursement only	**-12.0%**	**41.3%**	**112.7%**	*(CPT Revenue – Net Cost) / Net Cost*
ROI — with cost avoidance included	**-12.0%**	**440.0 %**	**862.7%**	*(CPT + Avoidance – Net Cost) / Net Cost*
Break-even patients (CPT only, no avoidance, no grant)	54	54	54	*Fixed costs / per-patient margin*
KEY INSIGHT FOR SWOT / EXECUTIVE SUMMARY				
CPT reimbursement alone produces modest positive ROI at 50+ patients, with break-even at ~54 patients. The transformative ROI comes from cost avoidance: preventing even one emergency air transport ($45,000) per year shifts clinic ROI from single digits to 300%+. For the SWOT, present both numbers: "CPT-only ROI of 6–42% at 50–80 patients; 300–480%+ when factoring in emergency transport avoidance."				
SOURCES				
CPT 2026 rates: Tenovi (tenovi.com/rpm-cpt-codes-2026); Prevounce (blog.prevounce.com)				
2026 CMS changes: Nixon Law Group (nixonlawgroup.com); CMS 2026 PFS Proposed Rule				
Air transport: American Action Forum 2023 ($36K median); CMS rural area 1.5x multiplier				
CHW wages: Bureau of Labor Statistics, Community Health Workers				
RHTP: $50B program FY2026–2030 (Babyscripts analysis; AHA; CMS)				
Yellow cells = adjustable assumptions (blue text). Black text = formulas.				
All KAIROS-specific costs (platform pricing, device cost, grant %) are synthetic/plausible estimates.				

CASE STUDY

When Every Minute Counts and the Nearest Hospital Is 90 Miles Away

How KAIROS Helped One Rural Mother Catch a Silent Complication Before It Became an Emergency

Keisha's Story

Keisha Williams, 31, lives in Phillips County, Arkansas — a community where the nearest hospital with an OB unit closed four years ago. When she delivered her second child at a regional medical center 87 miles from home, everything went smoothly. She was discharged two days later with the standard instructions: rest, watch for warning signs, and come back in six weeks for a checkup.

But Keisha knew from her first pregnancy that the weeks after delivery are anything but simple. After her first child, she'd experienced headaches and swelling that she'd dismissed as normal exhaustion. It wasn't until a routine visit weeks later that a nurse caught her dangerously elevated blood pressure. She was lucky.

This time, her OB enrolled her in KAIROS before she left the hospital.

The Problem: 42 Days Without Being Seen

For the 2.3 million women of reproductive age living in U.S. maternity care deserts, the postpartum period is the most dangerous gap in care. Between discharge and that first follow-up appointment, new mothers are largely on their own — often managing recovery, a newborn, and older children with no clinical touchpoint.

The numbers tell a stark story:

- **4 in 5** pregnancy-related deaths in the U.S. are preventable
- **Over 35%** of U.S. counties are classified as maternity care deserts
- **Mental health conditions** are now the leading cause of maternal death, contributing to nearly 1 in 4 pregnancy-related fatalities
- **Black women** are 3 to 4 times more likely to die from pregnancy-related complications than white women

For women like Keisha — living in a rural county with no OB unit, limited broadband, and a 90-minute drive to the nearest specialist — traditional remote monitoring apps aren't a realistic solution. Most require consistent Wi-Fi, a smartphone, and the kind of uninterrupted attention that's hard to find when you're recovering from birth with a toddler in the next room.

The Solution: Care That Works Where You Are

KAIROS was built for exactly this scenario. The name comes from the Greek word *kairos* (καιρός), meaning the right or critical moment — because in postpartum recovery, catching a complication early can be the difference between a phone call and an ambulance.

When Keisha was enrolled, she received a simple monitoring kit: a blood pressure cuff and a digital scale. No apps to download. No passwords to remember. No Wi-Fi required. A small cellular gateway device, about the size of a deck of cards, automatically sends her readings to her care team over basic cellular signal.

Her daily routine took less than three minutes:

1. Take her blood pressure
2. Step on the scale
3. The devices send the data automatically — no buttons, no syncing, no uploads
4. Her care team reviews the readings and reaches out if anything looks off

The Moment That Mattered

Nine days after delivery, Keisha felt fine. A little tired — but what new mother isn't? She took her morning readings as usual.

Her blood pressure was 158/102.

She didn't feel any symptoms. No headache, no blurred vision, no swelling. But the KAIROS system flagged it immediately. Because the device performs a risk check before transmission, the alert was prioritized even on Keisha's weak rural cellular connection. Within 20 minutes, a community health worker from her local clinic called her.

> *"I honestly felt normal. If I hadn't been taking those readings every morning, I never would have known something was wrong. They caught it before I even had a symptom."*
>
> — Keisha W., Phillips County, AR

Her provider adjusted her medication remotely. Over the next week, her readings trended back down. She never had to make the 90-mile drive to the hospital. She never had to bring her newborn and toddler to an emergency room. A potentially life-threatening complication was caught and managed from her kitchen table.

Why KAIROS Worked When Other Solutions Wouldn't

Keisha's experience highlights the specific barriers that KAIROS was designed to overcome:

- **No reliable internet?** KAIROS uses ultra-low-bandwidth SMS-based syncing and cellular transmission. It has achieved 98% data transmission success even in areas with minimal signal strength.
- **No smartphone?** The cellular gateway sends readings directly — no app, no downloads, no troubleshooting.
- **Limited digital comfort?** The devices are one-button operation. No passwords, no data entry, no technical setup.
- **Far from a specialist?** KAIROS connects patients to their care team in real time, reducing the need for long-distance travel during recovery. Projected average reduction: 450 miles of maternal travel per pregnancy episode.
- **No local OB?** KAIROS trains and integrates community health workers who already exist in rural communities, bridging the gap between patients and remote specialists.

For Providers and Health Systems: The Clinical and Financial Case

KAIROS isn't just a patient tool — it's a clinical infrastructure solution for providers and health systems operating in resource-constrained environments.

Clinical Impact

- **Earlier detection:** Comparable RPM programs have demonstrated a 13-day reduction in time to detect preeclampsia, the leading hypertensive complication in postpartum women.
- **Improved adherence:** Following UX optimization for low-signal environments, projected patient adherence rises from 42% to 78%.
- **Postpartum engagement:** Industry data shows RPM users are 2x more likely to attend postpartum follow-up visits and show 127% improved adherence at the 30-day mark.

Financial Impact

- **Cost avoidance:** A single emergency helicopter transport from a rural county to an urban Level III facility costs $45,000–$60,000. Each prevented emergency transfer represents direct savings.
- **RPM reimbursement:** CPT codes 99453 and 99454 enable providers to generate revenue from remote monitoring, with reimbursement typically exceeding the monthly per-patient software cost.
- **RHTP alignment:** KAIROS includes automated reporting for the $50 billion Rural Health Transformation Program (2026–2030), helping rural providers meet grant compliance requirements without additional administrative staff.
- **Estimated ROI: 145%** for rural clinics through combined CPT code capture and Transformation Incentives.

Designed for Rural and Safety-Net Settings

- **Works in the 52.4% of rural hospitals** that no longer offer obstetric services — acts as a virtual OB monitoring unit for general clinics.
- **Serves the 47% of rural births** covered by Medicaid, where the mortality rate is 3x higher than in urban commercially insured populations.
- **Deploys without EMR integration** — a standalone dashboard that meets RHTP sustainability goals without requiring Epic or Cerner infrastructure.

Essential Care for the Moments That Matter Most

Keisha's story isn't exceptional — it's typical. Across the country, new mothers in rural and underserved communities face the same gap: the weeks after birth when they're most vulnerable and least monitored. The technology to close that gap doesn't need to be complicated. It needs to be reliable, simple, and designed for real life — not ideal conditions.

KAIROS delivers exactly that. Essential care, wherever you are, for the critical moments after birth.

"I didn't need a fancy app. I needed someone watching out for me when I couldn't watch out for myself. That's what KAIROS did."

— Keisha W.

Sources & References

Maternal mortality and preventability statistics: CDC Maternal Mortality Review; National Center for Health Statistics.

Maternity care desert data: March of Dimes 2024 Report; Digital Health Insights.

Mental health as leading cause of maternal death: CDC, cited in Marani Health public materials.

Racial disparities in maternal mortality: Indiana University (2022); AfroTech reporting on Navigate Maternity.

13-day preeclampsia detection improvement: George Washington University / Babyscripts research.

Postpartum visit adherence (2x improvement, 127% at 30 days): Babyscripts published outcomes.

RPM ROI data and cost avoidance: Marani Health blog series; Babyscripts economics of digital health.

Rural broadband and connectivity barriers: Sciencedirect (2025), cited in team research document.

RHTP funding: Babyscripts policy analysis of the Rural Health Transformation Program.

KAIROS projected data (adherence improvement, data transmission rates, travel reduction, ROI): Synthetic/plausible estimates based on competitor benchmarks and market research.

Executive Summary: Strategic Redesign and Market Positioning for Kairos

To: Chief Product Officer

From: User Experience Research Team

Subject: Repositioning Kairos to Capture the Rural Health Transformation (RHT) Opportunity

Strategic Overview

The current maternal health market is dominated by "concierge-style" or "deep-integration" incumbents like Babyscripts and Marani Health. While these companies succeed in urban, high-bandwidth hospital environments, they leave a massive gap in the **35% of U.S. counties classified as maternity care deserts**. These regions often suffer from a "double desert" effect: a lack of obstetric providers compounded by **broadband internet deserts**, where households lack reliable 100/20 Mbps access.

Our redesign of Kairos focuses on "Kinetically Accessible Integrated Remote Obstetric Systems" (Kairos), shifting our product from a heavy, data-reliant app to a **lightweight, offline-first infrastructure** optimized for 3G/LTE smartphone networks and the specific funding incentives of the **$50 billion Rural Health Transformation Program (RHTP)**.

Market Gap Analysis (Derived from Competitor SWOT)

Our research into competitors identified three critical gaps Kairos will exploit:

1. **Technical Bloat:** Babyscripts and Marani require high-speed data for real-time syncing and high-resolution educational video content. **Kairos** will utilize "Ultra-Lean" data packets, reducing transfer sizes by **over 90%.**

2. **Implementation Friction:** Babyscripts relies on "white-glove" EMR integrations (Epic/Cerner) that rural clinics cannot afford to maintain. **Kairos** will deploy as a "Universal Standalone" dashboard that meets RHTP sustainability goals.

3. **Adherence vs. Access:** While Navigate Maternity focuses on social determinants (SDoH), it lacks the clinical automation to drive adherence in low-density populations.

Key Recommendations for Product Development

- **The "Offline-First" UX:** Redesign the Kairos app to cache all educational and screening content locally. Biometric data will be transmitted via **low-frequency SMS bursts** rather than persistent web-socket connections, ensuring **100% data ascertainment** even in low-signal areas.

- **RHTP-Incentivized Dashboard:** Develop an automated "RHTP Compliance Reporting" engine. This will allow rural providers to automatically generate the documentation required for RHTP grant renewals and value-based care payments without additional administrative staff.

- **Essential-Only RPM Kit:** Simplify the hardware kit to a ruggedized, cellular-linked blood pressure cuff and weight scale that functions without a smartphone, targeting the **15.4%** of rural residents who are currently uninsured and lack high-end mobile devices.